PROJECT GUIDE

Harmonic-Wise Paper Reimplementation Pipeline

Paper-Faithful Offline Reimplementation Guide

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This guide explains the repository-owned harmonic-wise pipeline that was added to reproduce the offline logic of `reference/RCIM_ML-compensation.pdf`.

This topic is not the same as the repository's direct-TE training families.

It belongs to the paper-faithful reproduction branch, which the backlog now calls `Track 1`.

The purpose of this branch is clear:

- recreate the paper's harmonic-wise modeling logic inside this repository;
- obtain an offline comparison protocol that is genuinely paper-aligned;
- prepare the bridge toward later online compensation work;
- keep the result inspectable for colleagues, reviewers, and future deployment work. At the current repository state, this guide covers the offline branch only. The online compensation loop and the final `Table 9` style benchmark are still future work.

Guide Export Folders

This topic separates its guide-local material into:

- `assets/`
  diagrams and `NotebookLM` source packages.
- `English/`
  reserved for future exported English presentation outputs.
- `Italiano/`
  reserved for future exported Italian presentation outputs.

Why This Pipeline Exists

The repository had already trained several strong direct-TE models.

Those models predict the TE curve directly from angular position plus operating conditions.

That is useful, but it is not how the reference paper is structured.

The paper works in a different way:

1. represent each TE curve through selected harmonics;
2. predict the harmonic quantities from operating conditions;
3. reconstruct the TE curve from the predicted harmonic stack;
4. later apply that reconstructed TE information to compensation logic.

So this pipeline exists because the repository needed a branch that answers a different question: can the paper's harmonic-wise logic be reproduced here, with repository-owned code, artifacts, and reports?

Track Placement

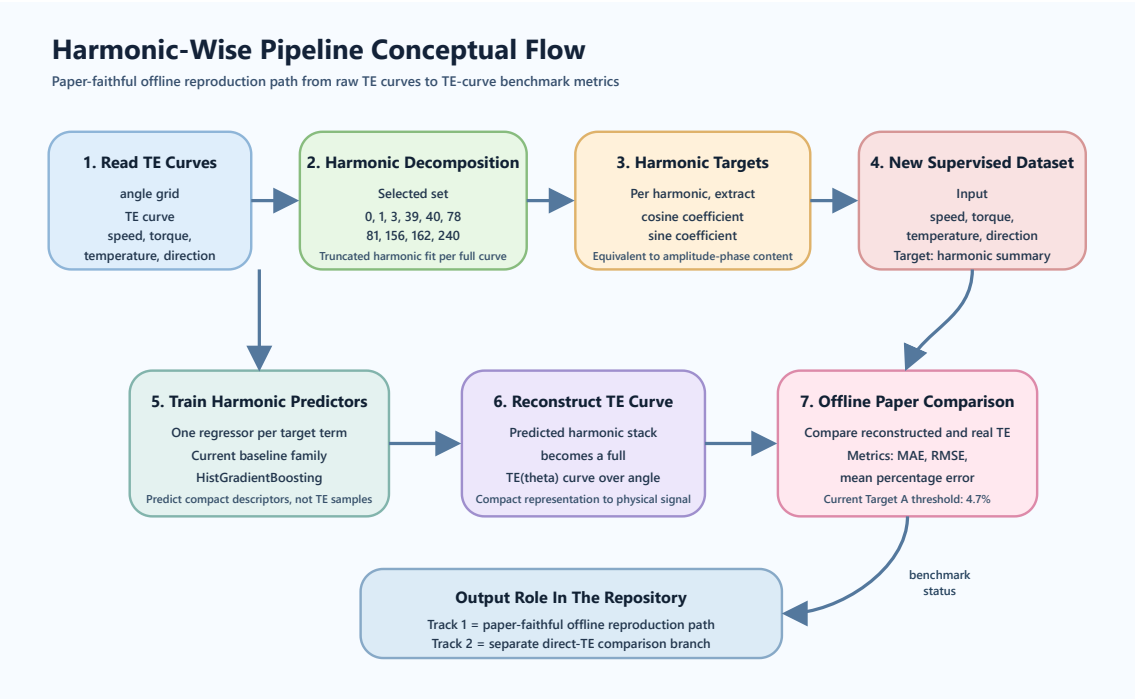
The repository now keeps two comparison tracks:

- `Track 1`

  paper-faithful harmonic-wise reproduction.
- `Track 2`

  repository direct-TE comparison under a common offline TE-curve evaluator. This guide is only about `Track 1`.

End-To-End Flow



Harmonic-wise pipeline conceptual flow

The pipeline starts from full TE curves, converts them into a harmonic summary, predicts those harmonic terms from operating conditions, and then reconstructs the TE curve for evaluation.

That means the model does not learn point-wise TE samples directly.

It learns a compact representation of the whole curve.

Stage 1. Read Curves And Operating Conditions

Each dataset item contains

- an angular grid;
- a TE curve on that angular grid;
- operating conditions such as `speed`, `torque`, `temperature`, and

  `direction`. In the direct-TE branch, the angle itself is part of the model input. In this harmonic-wise branch, angle is not the final learning input. Instead, the full angle-resolved TE curve is treated as a signal to summarize.

Stage 2. Decompose The TE Curve Into Selected Harmonics

The pipeline uses a selected harmonic set aligned with the paper-oriented comparison scope:

- `0`
- `1`
- `3`
- `39`
- `40`
- `78`
- `81`
- `156`
- `162`
- `240`

  These harmonics were chosen because they capture the practical periodic structure emphasized by the paper and by the repository's own benchmark notes. The decomposition stage fits a truncated harmonic representation of the TE curve. Conceptually, the curve is written as: `TE(theta) ~= a0 + sum_k [c_k cos(k theta) + s_k sin(k theta)]` where:

- `a0` is the offset term;
- `c_k` is the cosine coefficient for harmonic `k`;
- `s_k` is the sine coefficient for harmonic `k`.

Stage 3. Convert The Curve Into Harmonic Targets

For each selected harmonic, the current implementation stores the signal in coefficient form:

- cosine coefficient;
- sine coefficient.

  This form is convenient for stable regression and reconstruction. It is also directly connected to the paper quantities:

- amplitude `A_k`;
- phase `phi_k`.

  The relationship is:

- `A_k = sqrt(c_k^2 + s_k^2)`
- `phi_k = atan2(-s_k, c_k)` in the chosen convention

  So even though the repository currently predicts coefficient pairs, those coefficients still represent the same harmonic content that can be expressed as `A_k` and `phi_k`.

Stage 4. Build A New Supervised Dataset

This is the key conceptual shift.

The problem is no longer

- input = `angle + operating conditions`
- target = `TE(angle)`

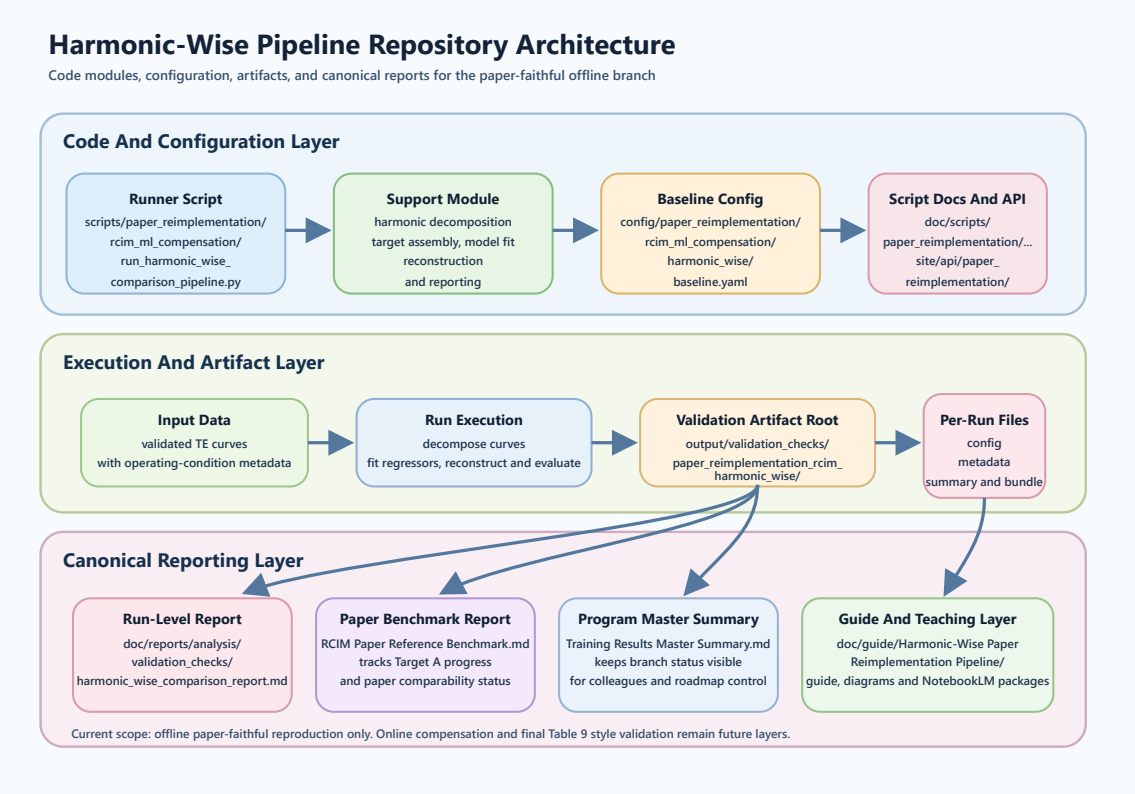
  The problem becomes:

- input = `speed + torque + temperature + direction`

- target = harmonic summary of the whole TE curve

  So each supervised sample corresponds to one operating-condition point and one whole-curve harmonic representation. That is why the pipeline is called harmonic-wise. It predicts the ingredients of the curve, not the point-wise curve samples.

Stage 5. Train Predictors On Harmonic Targets



Harmonic-wise repository architecture

The current repository baseline uses

- one regressor per harmonic target term;
- model family: `HistGradientBoosting` .

  That means the pipeline fits many small tabular regressors rather than one large end-to-end point-wise neural model. This is already directionally aligned with the paper, where tree and boosting families dominate the deployable harmonic stack. The baseline script is:

- `scripts/paper_reimplementation/rcim_ml_compensation/run_harmonic_wise_comparison_pipeline.py`

  The main support module is:

- `scripts/paper_reimplementation/rcim_ml_compensation/harmonic_wise_support.py`

  The current baseline configuration is:

- `config/paper_reimplementation/rcim_ml_compensation/harmonic_wise/baseline.yaml`

Stage 6. Reconstruct The TE Curve

Once the model has predicted the harmonic coefficients for a new operating point, the pipeline rebuilds the full TE curve over angle.

This is the bridge between

- compact harmonic prediction;
- physical TE behavior over the full angular cycle.

  Without reconstruction, the pipeline would only output abstract coefficients. With reconstruction, it becomes comparable to the paper's offline TE-curve validation logic.

Stage 7. Evaluate The Reconstructed Curve

The reconstructed TE curve is compared against the real TE curve on held-out validation and test splits.

The main metrics are

- `MAE`
- `RMSE`
- mean percentage error

  The mean percentage error over the full TE curve is the key paper-comparable metric for the current offline phase. The first repository-owned baseline established that protocol and produced:

- validation mean percentage error: `9.474%`
- test mean percentage error: `9.403%`

  The current offline paper target is:

- `<= 4.7%`

  So the comparison protocol now exists, but `Target A` is not closed yet.

Why The First Baseline Still Matters

Even though the first baseline does not yet match the paper threshold, it already proved something important.

The truncation-only oracle with the same harmonic set reaches a much lower error than the learned predictor.

That means

- the selected harmonic set is expressive enough;
- the bottleneck is the operating-condition predictor;
- the next improvement work should focus on the predictor and target design,

  not on abandoning the harmonic representation itself.

Offline Playback Probes

The current pipeline also includes offline `Robot` and `Cycloidal` style motion-profile playback probes.

These are not yet the final online compensation benchmark.

They are repository-owned preparation steps that

- exercise the predicted harmonic stack on profile-like conditions;
- prepare the path toward later online evaluation;
- keep the eventual `Table 9` target visible from the beginning.

What This Pipeline Is Not

This pipeline is not

- the direct-TE training branch;
- a finished online compensation system;
- a TwinCAT deployment package;
- the final `Table 9` benchmark.

  It is an intermediate but essential branch between completed `Wave 1` and the later online compensation path.

Current Repository Outputs

The current validation artifacts live under

- `output/validation_checks/paper_reimplementation_rcim_harmonic_wise/`

  The current human-readable validation report is generated under:
- `doc/reports/analysis/validation_checks/`

  The canonical comparison reports that summarize the branch are:
- `doc/reports/analysis/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`

Practical Reading Map

If you want to understand the branch quickly, read in this order:

1. this guide;
2. `doc/reports/analysis/RCIM Paper Reference Benchmark.md`;
3. `doc/reports/analysis/validation_checks/...harmonic_wise_comparison_report.md`;
4. the pipeline script and support module.

Summary

The harmonic-wise paper-reimplementation pipeline changes the learning problem from:

- point-wise TE prediction

  to:
- operating-condition-to-harmonic prediction

  and then reconstructs the TE curve for evaluation. That is exactly why it matters. It gives the repository a paper-faithful offline benchmark path instead of only a direct-TE leaderboard.

Related Reading

- `doc/reports/analysis/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`
- `doc/guide/project_usage_guide.md`